

Ariv Vaidya

Student | Science Enthusiast | Volunteer | Leader

Profile

Driven and curious high school junior passionate about biomedical research and computational biology. Successfully contributed to neurodegenerative disease research at Harvard Medical School and Mass General Brigham, developing predictive models using R and Python, and presenting findings at major conferences. Proven leader and varsity tennis captain with a strong drive for excellence on and off the court. Grounded in empathy through mentoring underserved students and performing classical guitar at senior living centers to uplift and connect.

Education

British International School of Boston

JANUARY 2022 – JANUARY 2026

- Enrolled in International General Certificate of Secondary Education (IGCSE) program in Freshman and Sophomore year. Maintained 4.5 GPA; enrolled in accelerated mathematics; electives: Business and Music
- Enrolled In International Baccalaureate Diploma Program (IBDP) in junior and senior year. Maintained 45 Points with higher level: Biology, Chemistry, AI Math. Standard level: English, Spanish, ESS

Experience

Research Trainee at Khurana Lab for Neurology, Division of Movement Disorders, Harvard Medical School and Massachusetts General Brigham, Boston, Massachusetts, United States

JUNE 2024 – PRESENT

- Conducted advanced data analysis using R and Python on the Harvard Biomarker Study, involving 4000 patients, to cluster progression rates and predict clinical outcomes based on baseline characteristics.
- Collaborated on cutting-edge research, including the Seed Amplification Assays (SAA) to study protein aggregation in neurodegenerative diseases.
- Gained comprehensive exposure to clinical neurology through shadowing neurologists, attending lab meetings, and participating in seminars on neurodegenerative diseases at Brigham and Women's Hospital.

Data Science Intern, Mediar Therapeutics, Cambridge, Massachusetts, United States

JUNE 2023 – AUGUST 2023

- Served as a data scientist intern at Mediar Therapeutics, a clinical-stage biotechnology company, which develops therapies for lung, liver, and kidney fibrosis.
- Developed an R-based application for sorting and visualizing large RNA sequencing datasets, enhancing experimental analysis of gene expression for scientific teams.
- Participated in lab meetings, journal clubs, and seminars focused deepening my understanding of fibrosis research, therapeutic strategies and the drug development pipeline

Programmer, Green Pyramid Energy, Pune, Maharashtra, India

SEPTEMBER 2022 – JANUARY 2023

- Contributed to IoT-based water management solutions at Green Pyramid Energy, a leader in developing wireless equipment for water monitoring and control using advanced technologies.

Details

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Links

[LinkedIn](#)

Skills

Machine Learning and AI

Data Analysis and Visualization

Scientific Research and Experimentation

Leadership and Team Collaboration

Public Speaking and Presentation

Mentorship and Teaching

Community Engagement

Time Management and Multitasking

Languages

English

Marathi

- Developed a preliminary program to calculate water depth in storage tanks using sound frequency analysis, enhancing accuracy in water level measurement.
- Created a program and conceptual framework for image processing in robotics, focusing on removing blurred images and duplicates from training datasets to improve machine learning model performance.

Leadership

Class President at British International School of Boston

MAY 2025 – PRESENT

- Elected Class President, leading student initiatives, representing classmates' interests, and fostering a positive, inclusive school community through effective communication and collaborative leadership

Founder of Current Events Club at British International School of Boston

SEPTEMBER 2024 – PRESENT

- Founded and led the Current Events Club, creating a collaborative platform for middle and high school students to develop weekly news articles for school newspapers.
- Mentored and guided students in journalistic writing, critical analysis, and current events reporting, fostering skills in communication and media literacy.
- Developed a structured program that encourages student engagement with contemporary issues ranging from politics, sports, medicine, science and the environment and published the articles on the school's newspaper

Science and MIT Ambassador at British International School of Boston

SEPTEMBER 2022 – PRESENT

- Served as Science Ambassador for BISB at MIT, managing events, teaching experiments to younger students, writing MIT blogs, and supporting scientific outreach.
- Helped organize the Cambridge Science Festival, advancing MIT's mission to make science accessible and inspire future innovators.
- Invited to MIT to trial and evaluate new Nord Anglia educational projects, providing feedback on innovative learning methods and technologies.

Peer Mentor and Advisor at British International School of Boston

SEPTEMBER 2022 – PRESENT

- Supported middle school students as a peer mentor by helping them navigate academic and social challenges, offering both academic assistance and emotional support, fostering friendships, and leading team-building activities to promote a positive and inclusive school environment.

Open House Representative at British International School of Boston,

SEPTEMBER 2022

- Served as an Open House Representative, providing tours of the school and engaging Q&A sessions for prospective students and families to represent the school

Head of Bulldog Inquirer Newspaper at British International School of Boston

MAY 2025 – PRESENT

- Led the Bulldog Inquirer as head of the school newspaper, overseeing content creation, managing the editorial team, editing articles, and coordinating publication to inform and engage the student body

Varsity Tennis Player and Captain at British International School of Boston

SEPTEMBER 2022

- Demonstrated leadership and athletic excellence by guiding the high school tennis team, balancing competitive performance with team development and strategic play.

Community Service

STEM Student Volunteer at Alexander Twilight Academy

JANUARY 2025 – PRESENT

- Served as a Teacher Assistant for Math at Alexander Twilight Academy, a free academic catalyst program supporting high-potential students from under-resourced backgrounds in the Boston area.
- Contributed to ATA's mission of closing the achievement gap by providing rigorous academic support for middle school students

Student Volunteer at Volley Against Violence,

OCTOBER 2024 – PRESENT

- Volunteered in a national initiative to foster community engagement by helping organize tennis sessions where police officers interact with local children, promoting positive relationships.
- Supervised and assisted coaches in practice sessions, focusing on skill development and meaningful interactions between youth and Boston police.

Extracurricular Activities

Classical Guitar Player

JUNE 2015 – PRESENT

- Played over fifteen recitals at Brookline Music School
- Volunteered at senior living centers in Boston to entertain residents with guitar music sessions
- Awards:
 - 01/2025: Awarded the Best Musician of the Year in Highschool
 - 04/2024: Awarded 3rd place in Nord Angelia (NaE) global music competition
 - 01/2024: Awarded the Best Musician of the Year in Highschool
 - 05/2023: Won gold medal in the Honors Evaluation at Brookline Music School in the string's competition.

Tennis Player

AUGUST 2020

- Played competitive tennis with a Universal Tennis Rating (UTR) of 5.38; ranked top 50 in New England

Programing

OCTOBER 2019

- Started taking classes at Coding Butterfly where I learned different programing paradigms, pseudo code, and built problem solving skills in python
- Took classes in The Advanced Computer Science Academy, where I learned iterative and recursive solutions, implemented mathematical algorithms and advanced object-oriented programming such as manipulating classes

Super-curricular Activities

Attendee and Poster Presenter at World Pre-Health Conference

MARCH 2025 – APRIL 2025

- Attended the World Pre-Health Conference at Harvard, engaging with keynote speakers on novel medical research, participating in lectures on medical school applications, and gaining hands-on experience in suturing and surgical precision alongside a diverse group of aspiring healthcare professionals.
- Presented my research poster at the World Pre-Health Conference symposium, sharing findings with professors, undergraduates, and peers

R Programming Language Course

JULY 2023 – AUGUST 2023

- Completed 4 courses specialization in R. Learned basic commands, manipulation of data, visualization of data with different platforms and creating a User Interface (UI) in R

Reading Books Related to Biology, Medicine and Computational Sciences

JANUARY 2022 – PRESENT

- *Genome* by Matt Ridley; *The Fragile Earth* by David Remnick and Henry Finder; *The Checklist Manifesto* and *Complications* by Atul Gawande; *AI Revolution in Medicine* by Peter Lee, Carey Goldberg, and Isaac Kohane

Awards and Acknowledgements

3rd Place in Boston Massachusetts State Science Fair,

APRIL 2025

Alumni, National Science Fair, Broadcom Masters,

SEPTEMBER 2022

First Place, Boston Region V STEM Middle School Science Fair,

MAY 2022

Honorable Mentions, Massachusetts Middle School Science and Engineering Fair,

MAY 2022

First Place Science Fair Competition, Heath School,

MAY 2021

First Place Science Fair Competition, Heath School,

JUNE 2020